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# Chapter 10

# Introduction to Switching

We have come to the point in the book where we are going to switch directions (pardon the pun) and move away from routing as a topic and focus more on a network switch. This chapter is designed to test your knowledge of the fundamentals of network switches. This chapter is acting as more of a primer for the next chapter because it tests you on topics such as the purpose of a switch, the flow of communication, and basic configuration of the switch.

## The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding the fundamentals of a Cisco switch
- Understanding the pathway to communication and addressing
- Looking at basic switch configurations, such as IP address and gateway assignment

## What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- A switch is a layer-2 device that filters traffic based off the destination MAC address. The switch uses a MAC address table, which lists the MAC address of the systems connected and what port those systems are connected to. The switch can populate the MAC address table dynamically by looking at the source address of a received frame. Administrators can manually add entries to the MAC address table as well.
- The mode button on the front of the switch toggles the display mode, which changes the meaning of the LEDs on the front of the switch. For example, if you are in status display mode, a blinking light means traffic is being transmitted on that port. But if the display mode is set to speed, then a blinking light on a port means you are running at 1 Gbps.
- Know the pathway of communication for the exams! When a message is sent between two systems, the source and destination IP addresses in the message stay the same each step of the way, but the source and destination MAC addresses change each step of the way.

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## Introduction to Cisco Switches

***352–359*** Choose the best answer(s).

***352.*** Bob calls complaining that he cannot access the network. When you look at the switch, you notice the port he is connected to is displaying an amber light. What does this mean?

(A) The IP address on the switch is incorrect.

(B) The default gateway setting on the switch is incorrect.

(C) The default gateway setting on Bob's station is incorrect.

(D) The port has been disabled.

***353.*** After powering on the Cisco switch you notice that the System LED is green. What does this indicate?

(A) The system started without problems and is operational.

(B) There is a power problem.

(C) There is a problem with the default VLAN.

(D) There were POST errors.

***354.*** The switch has the display mode set to Status and port 4 is blinking green. What does this indicate?

(A) There is a link

(B) Waiting for traffic to be sent

(C) The port is running in full duplex

(D) Data is being transmitted

***355.*** After powering on the Cisco switch, you notice that the System LED is amber. What does this indicate?

(A) The system started without problems and is operational.

(B) There is a power problem.

(C) There is a problem with the default VLAN.

(D) There were POST errors.

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**356.** You wish to verify which ports on the switch are running in full duplex mode. How would you do this?

(A) Switch the display mode to duplex and watch for a solid green light on the different ports.

(B) Switch the display mode to duplex and watch for an amber light on the different ports.

(C) Switch the display mode to duplex and watch for no light on the different ports.

(D) Switch the display mode to duplex and watch for a flashing green light on the different ports.

**357.** When your switch is set to the display mode of speed, what does the LED for a 1 Gbps port look like?

(A) Flashing amber

(B) Solid amber

(C) Solid green

(D) Flashing green

**358.** The switch display mode is set to duplex and you have a computer connected to port number 4 with no light displaying on the port. What is the problem?

(A) There is no link.

(B) No traffic is being sent.

(C) The port is running in half duplex.

(D) The port is running in full duplex.

**359.** You have the display mode set to speed on your 10/100/1000 switch. Three of the ports are not displaying a light. What does this indicate?

(A) Ports are running at 1000 Mbps.

(B) Ports are running at 100 Mbps.

(C) Ports are running at 10 Mbps.

(D) No data is being transmitted.

## Understanding Switch Functionality

***360–378*** Choose the best answer(s).

**360.** A switch is an example of a layer-___ device.

(A) 1

(B) 2

(C) 3

(D) 4

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**361.** Which of the following protocols does a Cisco switch use for loop avoidance?

(A) CDP

(B) VTP

(C) STP

(D) PDC

**362.** When a switch receives a frame, what address does it look to in order to determine where to forward the frame?

(A) The destination MAC address

(B) The source MAC address

(C) The destination IP address

(D) The source IP address

**363.** Which of the following descriptions matches a collision domain?

(A) A group of systems that can have their data collide with one another

(B) A group of systems that receive one another's broadcast messages

(C) A group of systems between two firewalls

(D) A system that filters traffic by source IP address

**364.** Which of the following are considered core services offered by a switch? (Select three.)

(A) Filtering and forwarding

(b) Loop avoidance

(C) Address translation

(D) IP address assignment

(E) Address learning

(F) Name resolution

**365.** Which of the following descriptions matches a broadcast domain?

(A) A group of systems that receive one another's broadcast messages

(B) The area between two firewalls

(C) A group of systems that can have their data collide with one another

(D) A network device that shares a single IP address via port address translation

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**366.** Looking at the figure below, what does the switch do when it receives a frame destined for the MAC address of 00d0.bc8a.2766?

```
Switch#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0009.7c57.5674    DYNAMIC     Fa0/2
   1    0010.11d9.d001    DYNAMIC     Fa0/1
   1    00d0.bc8a.2766    DYNAMIC     Fa0/12
```

(A) It forwards the frame to port 1.

(B) It forwards the frame to port 2.

(C) It will flood the frame.

(D) It forwards the frame to port 12.

**367.** What does a switch do with a frame it receives that is destined for a MAC address that is not stored in the MAC address table?

(A) Broadcasts the frame

(B) Floods the frame

(C) Sends the frame to the address of FF-FF-FF-FF-FF-FF

(D) Stores it until the destination MAC is in the MAC address table

**368.** Your manager is wondering what the benefit of switches over hubs are. What would you say?

(A) Hubs filter traffic and only send the traffic to the destination port, while a switch sends all traffic to all ports.

(B) Switches filter traffic and only send the traffic to the destination port, while a hub sends all traffic to all ports.

(C) Hubs segment the traffic.

(D) Switches use a single collision domain for all ports.

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**369.** How many collision domains exist on a 24 port hub versus a 24 port switch?

(A) Hub – 1 collision domain / switch – 24 collision domains

(B) Hub – 24 collision domain / switch – 1 collision domains

(C) Hub – 48 collision domain / switch – 24 collision domains

(D) Hub – 24 collision domain / switch – 48 collision domains

**370.** Which of the following does the switch use to populate the MAC address table on the switch?

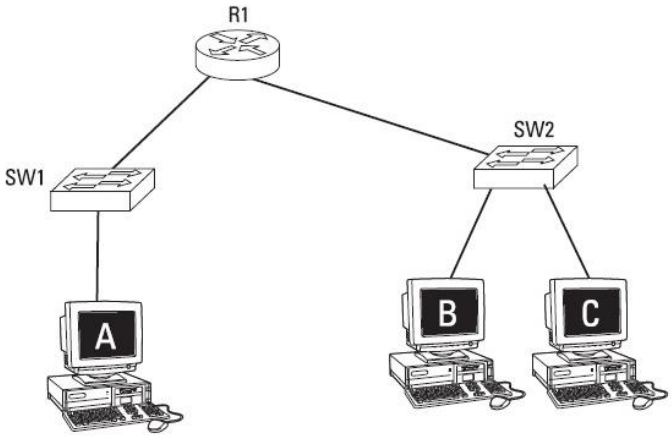
(A) The source MAC address of the frame

(B) The destination MAC address of the frame

(C) The source IP address of the frame

(D) The destination IP address of the frame

**371.** Looking at the figure below, how many collision domains and broadcast domains are there?




(A) 2 broadcast domains, 5 collision domains

(B) 2 collision domains, 4 broadcast domains

(C) 1 broadcast domain, 5 collision domains

(D) 1 broadcast domain, 4 collision domains

**372.** You have three switches connected together with crossover cables. How many broadcast domains exist?

(A) 3

(B) 1

(C) 6

(D) 2

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**373.** You have linked two switches together with a straight-through cable but cannot seem to communicate between the two switches. What is the solution?

(A) Use a console cable.

(B) Connect a router between the two switches.

(C) Use a crossover cable.

(D) Connect a bridge between the two switches.

**374.** Which switch operation mode waits to receive the first 64 bytes of the frame before forwarding the frame on to its destination?

(A) Store-and-forward

(B) Cache and deliver

(C) Fragment-free

(D) Cut-through

**375.** What is the general term for data received and processed by a switch?

(A) Frame

(B) Packet

(C) Segment

(D) Payload

**376.** Which switch operation mode waits until the entire frame is received before forwarding the frame on to its destination?

(A) Store-and-forward

(B) Cache and deliver

(C) Fragment-free

(D) Cut-through

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**377.** Looking at the figure below, what does the switch do when it receives a frame destined for the MAC address of 00d0.bc8a.2788?

```
Switch#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0009.7c57.5674    DYNAMIC     Fa0/2
   1    0010.11d9.d001    DYNAMIC     Fa0/1
   1    00d0.bc8a.2766    DYNAMIC     Fa0/12
```

(A) It forwards the frame to port 1.

(B) It forwards the frame to port 2.

(C) It will flood the frame.

(D) It forwards the frame to port 12.

**378.** Which switch operation mode starts forwarding the frame off to its destination as soon as the destination MAC address is received?

(A) Store-and-forward

(B) Cache and deliver

(C) Fragment-free

(D) Cut-through

## Understanding Data Flow

***379–381*** Choose the best answer(s).

**379.** When a packet is sent from a system on your network to a system on another network, which of the following is true of the packet as it is passed to your router from your system?

(A) Destination IP Address: your_router / Destination MAC Address: your_router

(B) Destination IP Address: the_remote_system / Destination MAC Address: your_router

(C) Destination IP Address: your_router / Destination MAC Address: the_remote_system

(D) Destination IP Address: the_remote_system / Destination MAC Address: the_remote_system

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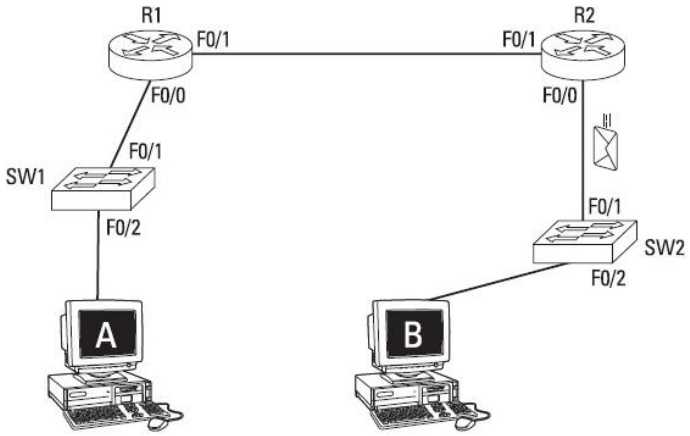
**380.** Computer A is sending a message to computer B. Looking at the figure below, which of the following are true of the message as it travels between R2 and SW2? (Select three.)




(A) The destination IP address is that of router R2.

(B) The destination MAC is that of Computer B.

(C) The source MAC address is that of router R1.

(D) The source MAC address is that of Computer A.

(E) The destination IP address is that of Computer B.

(F) The destination MAC is that of SW2.

(G) The source MAC is that of R2.

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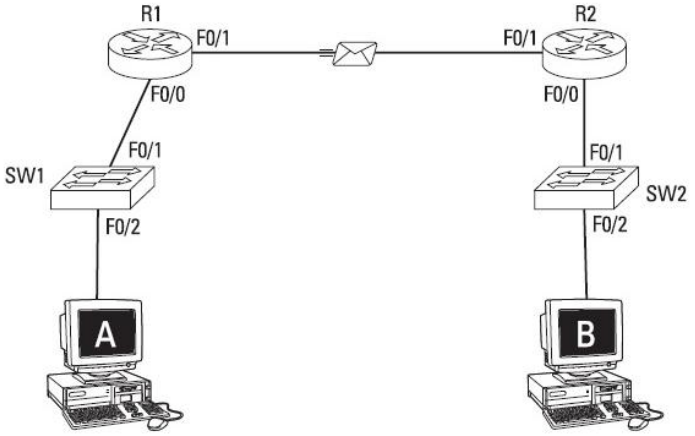
**381.** Computer A is sending a message to computer B. Looking at the figure below, which of the following are true of the message as it travels between R1 and R2? (Select three.)




(A) The destination IP address is that of router R2.

(B) The source IP address is that of Computer A.

(C) The source MAC address is that of router R1.

(D) The source MAC address is that of Computer A.

(E) The destination IP address is that of Computer B.

(F) The destination MAC is that of Computer B.

## Basic Switch Configuration

***382–389*** Choose the best answer(s).

**382.** Which of the following commands is used to change the name of the switch?

(A) Switch#`hostname NY-SW1`

(B) Switch(config)#`hostname NY-SW1`

(C) Switch(config)#`name NY-SW1`

(D) Switch#`name NY-SW1`

**383.** You wish to display your MAC address table. What command do you use?

(A) Switch(config)# `show mac-address-table`

(B) Switch(config-if)# `show mac-address-table`

(C) Switch# `view mac-address-table`

(D) Switch# `show mac-address-table`

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**384.** Your manager has asked you to telnet into one of the switches in the Las Vegas office, but you cannot seem to connect. You can telnet into the switch when you are in the Las Vegas office, but you cannot seem to telnet when you are on a remote network. What could be the problem?

(A) No Telnet password assigned to the switch

(B) No default gateway address assigned on the switch

(C) No default gateway address assigned on your router

(D) No Telnet password assigned on your router

**385.** Which of the following commands are used to assign an IP address to the switch?

(A) Use the following commands:

NY-SW1(config)#`interface f0/1`

NY-SW1(config-if)#`ip address 23.0.0.25 255.0.0.0`

(B) Use the following commands:

NY-SW1(config)#`ip address 23.0.0.25 255.0.0.0`

(C) Use the following commands:

NY-SW1(config)#`interface vlan1`

NY-SW1(config-if)#`ip address 23.0.0.25 255.0.0.0`

(D) Use the following commands:

NY-SW1(config)#`line vty 0 4`

NY-SW1(config-if)#`ip address 23.0.0.25 255.0.0.0`

**386.** Which of the following commands assigns a default gateway address to your switch?

(A) Switch(config-if)# `ip default-gateway 24.0.0.1`

(B) Switch# `ip default-gateway 24.0.0.1`

(C) Switch(config)# `ip default-gateway 24.0.0.1`

(D) Switch(config)#`default-gateway 24.0.0.1`

**387.** How long does a dynamic entry stay in the MAC address table?

(A) 300 seconds from time added to table

(B) 60 seconds from last time used

(C) 60 seconds from time added to table

(D) 300 seconds from last time used

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**388.** While looking at the figure below, what does the type of dynamic mean?

```
Switch#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0009.7c57.5674    DYNAMIC     Fa0/2
   1    0010.11d9.d001    DYNAMIC     Fa0/1
   1    00d0.bc8a.2766    DYNAMIC     Fa0/12
```

(A) The administrator has programmed the address.

(B) CDP has populated the address.

(C) The switch has learned the address based off traffic received.

(D) RIP has added the address.

**389.** You would like to change the amount of time an entry stays in the MAC address table on the switch to 400. What command would you use?

(A) `mac-address-table 400`

(B) `aging-time 400`

(C) `aging-time set-table 400`

(D) `mac-address-table aging-time 400`